



GUI-Based Face Recognition Login System with 2D Shooter Game

This Java project uniquely blends security and entertainment with a GUI-based face recognition login system coupled with an engaging 2D shooter game. The system offers a streamlined user experience by utilizing cutting-edge technologies to ensure secure access, followed by interactive gameplay. Throughout this presentation, we will explore the project overview, core features, technologies implemented, system design, and more, providing a full understanding of this innovative application.

Project Overview

Secure Login using Face Recognition

Ensures only authorized users gain access through biometric verification.

Fun 2D Shooter Game after Login

Delivers interactive entertainment immediately post-authentication.

Combines Security & Entertainment

Balances protection with engaging user experience.

User-Friendly GUI Experience

Clean and intuitive design for seamless navigation.



Main Features

Face Detection & Recognition

Real-time identification utilizing advanced image processing.

User Registration & Profile Management

Users can register, update profiles, and manage their data securely.

2D Shooter Game with Levels & Scores

Multiple levels offer increasing difficulty and competitive scoring.

Real-time Feedback & Notifications

Instant system responses to user actions improve engagement.

Secure Data Storage

User information and game data are safely stored using encryption.

Technologies Used

Java

Main programming language providing a robust backend.

JavaFX

Framework for building modern, responsive graphical user interfaces.

OpenCV

Library for efficient face detection and recognition workflows.

XML

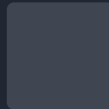
Used for structured and encrypted user data storage.

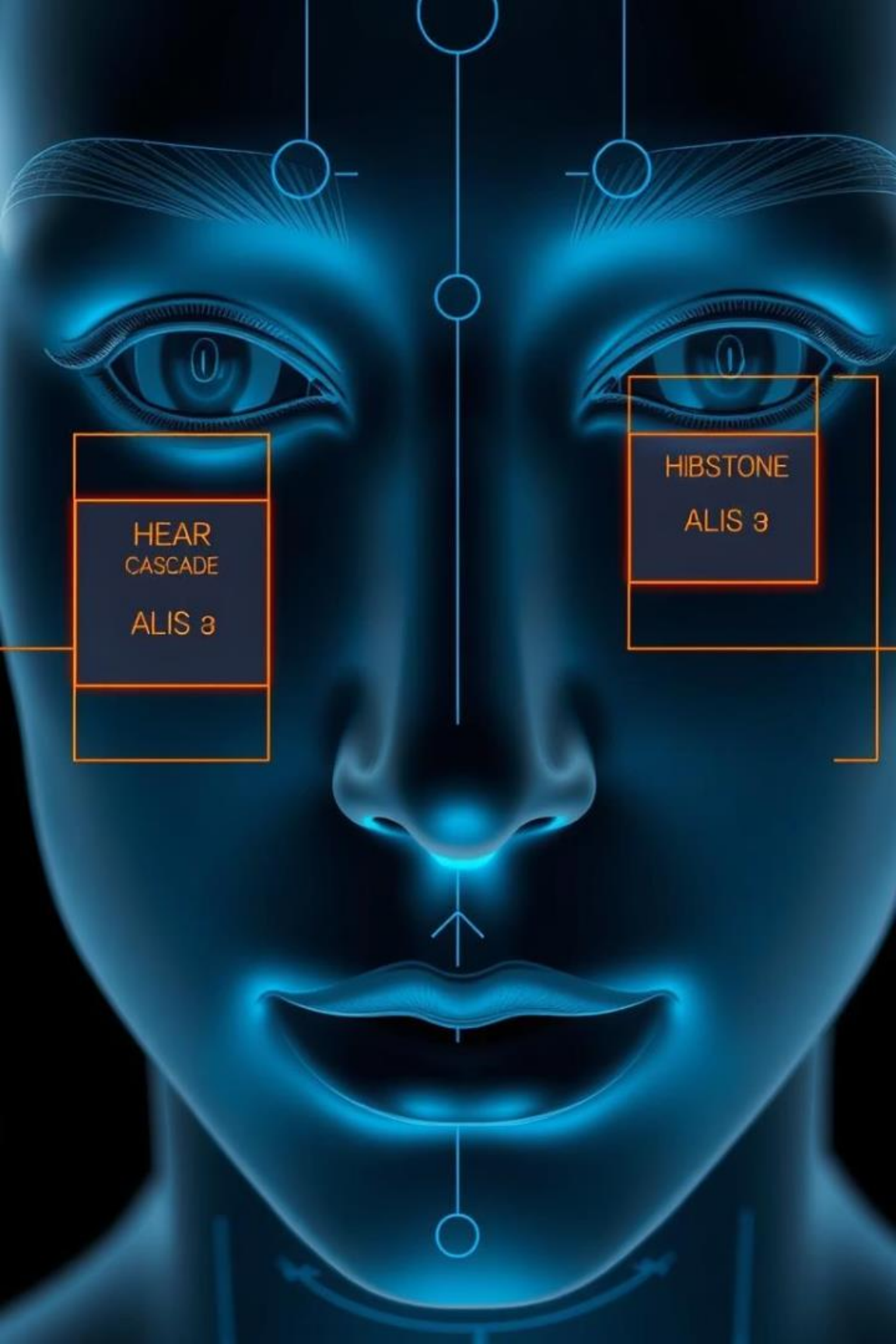
IntelliJ IDEA

Integrated Development Environment supporting Java and related tools.



System Design

-  **Modular Architecture**
System layers divided to enhance development and maintenance.
-  **Game Module**
Built with JavaFX Canvas offering smooth 2D graphics and controls.
-  **Face Recognition Module**
Handles biometric processing via OpenCV integration.
-  **Database Module**
Manages encrypted user data stored securely in XML format.



Face Recognition



OpenCV Haar Cascade Classifier

Efficient algorithm detecting facial features accurately in real-time.



Encrypted Data Storage

User face data is securely stored in encrypted XML files.



High Accuracy & Fast Processing

Optimized for quick and reliable user authentication.



Continuous Learning

Adaptive model improving face recognition over time.

2D Shooter Game

Simple & Addictive Gameplay

Easy to learn controls with engaging mechanics for players of all skill levels.

Multiple Levels & Difficulty

Increasing challenges to keep the game entertaining and rewarding.

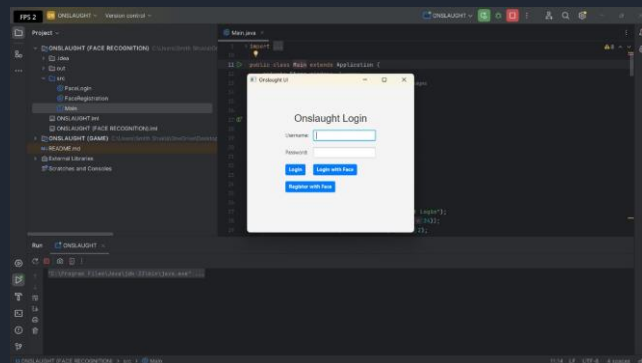
Score Tracking & Leaderboard

Competitive element motivates users to improve and compare results.

Dynamic Enemy AI

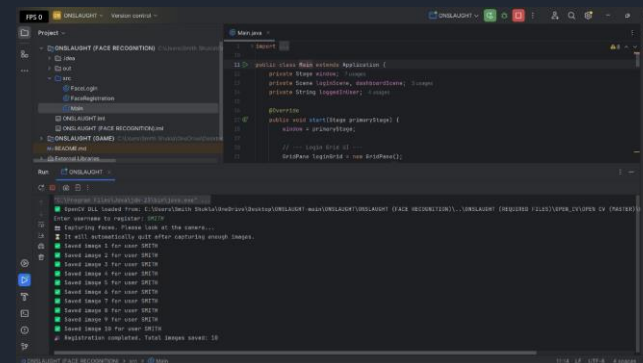
Enemies adapt their behavior for unexpected challenges during gameplay.

Screenshots



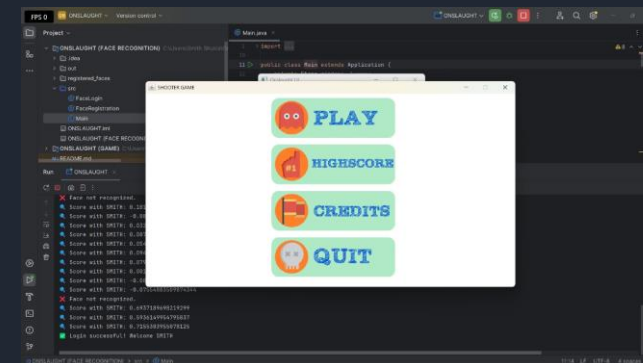
Login Screen

Clear interface with camera view for face scanning.



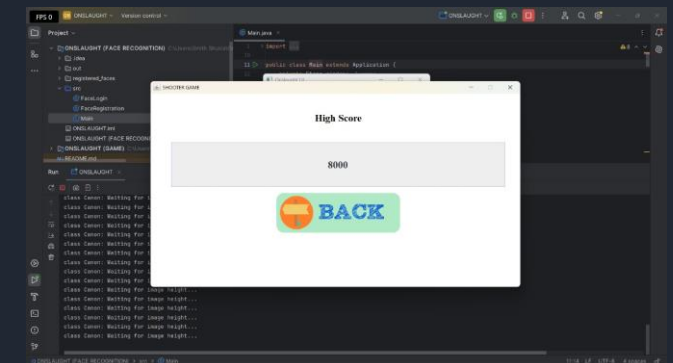
Registration Form

Secure and simple new user signup process.



In-Game Interface

Gameplay screen with health bars and score display.



High score

High scores to engage players.



Challenges & Solutions

Lighting Condition Variability

Implemented Adaptive Histogram Equalization to ensure face detection remains robust under varying lighting.

Real-time Processing Performance

Optimized algorithms to maintain quick recognition without lag impacting user experience.

Security Vulnerabilities

Secured face data and user info through strong encryption protocols and safe storage.



Future Work

Enhance Game Features & Levels

Add new gameplay mechanics, levels, and rewards for richer player engagement.

Integrate Cloud Storage

Allow remote storage of user profiles and game progress for multi-device access.

Improve Face Recognition Accuracy

Leverage deep learning techniques to increase precision and reduce false matches.

Mobile Application Development

Expand accessibility with dedicated apps for iOS and Android platforms.